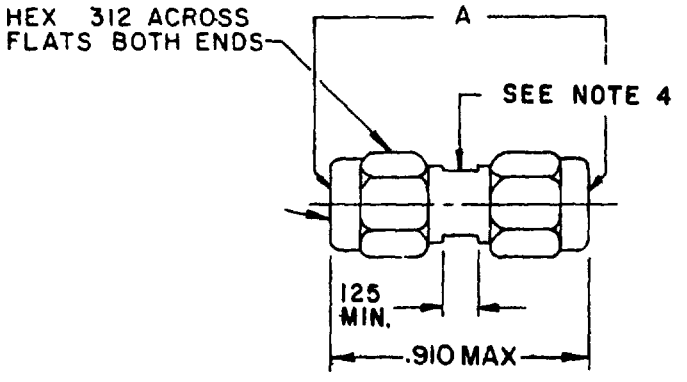
Ⓐ

MILITARY SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, IN-LINE,
(WITHIN SERIES SMA PLUG TO SMA PLUG), CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapter described herein shall consist of this document and the latest issue of Specification MIL-A-55339.



Reference	Series	Contact	Figure
A	SMA	Pin	2

INCHES	MM
.125	3.18
.312	7.92
.910	23.11

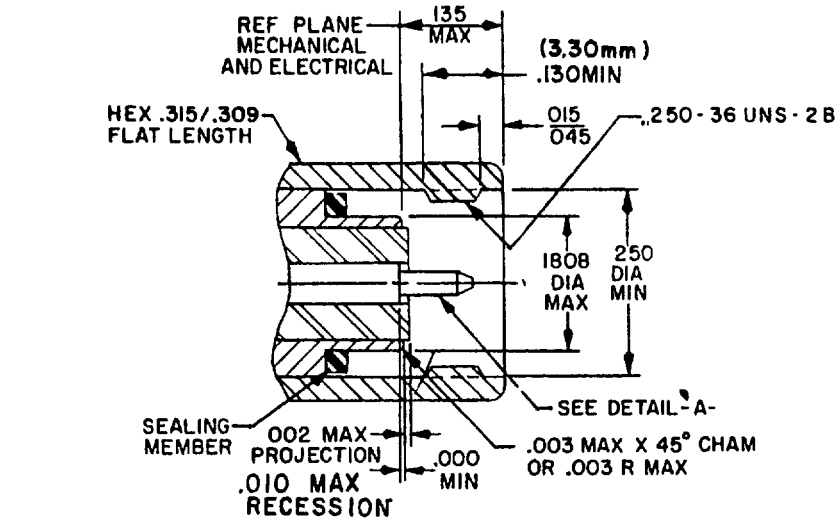
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. All undimensioned pictorial representations are for reference purposes only.
Ⓐ 4. Wrench flats are to accomodate standard wrench opening per H-28, app 10.

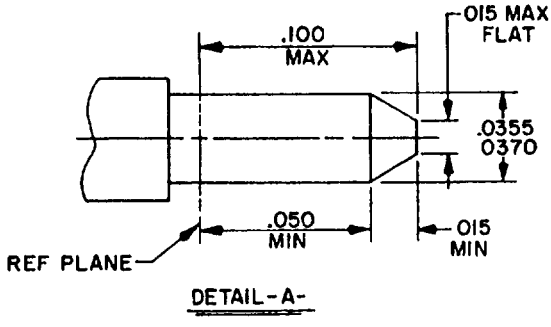
FIGURE 1. General configuration.

Ⓐ denotes changes

FSC 5935

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INCHES	MM
.002	.05
.003	.08
.015	.38
.0355	.90
.0370	.94
.045	1.14
.050	1.27
.100	2.54
.125	3.18
.135	3.43
.1808	4.59
.250	6.35
.309	7.85
.315	8.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Three holes .0168 (.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on coupling nut optional
Ⓐ 4. .150 minimum across the narrowest part of flat.

FIGURE 2. Mating dimensions for pin terminations

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DESIGN AND CONSTRUCTION:

General configuration: See figure 1.

Impedance 50 ohms, nom.

Working voltage Sea level - 335 Vrms.
70,000 feet - 85 Vrms.

Frequency range 0 to 18.0 GHz.

Temperature range: -65° to +165°C

PERFORMANCE

Dimensions: See figures 1 and 2.

Center contact retention: Axial force - 6 lb, min.
Torque - Not applicable.

Force to engagement and disengage: Longitudinal force - Not applicable.
Torque - 2 in. lb, max.

Coupling proof torque: 15 in, lbs, min.

Mating characteristics. Not applicable.

Permeability: <2.0

Seal:
Pressurized - Not applicable.
Weatherproof - Not applicable.

Insulation resistance: 5,000 megohms, min.

(A) VSWR: 1.10 + .008 F(GHz) at .5 to 18.0 GHz

(A) RF leakage (total): -60 dB, min 2 to 3 GHz.

RF insertion loss: .2 dB, max, between 2 to 3 GHz, min
(.06 $\sqrt{F(GHz)}$ dB max tested at 6 GHz)

Durability 500 cycles minimum at 12 cycles/min maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Dielectric withstanding: Test voltage - 1,500 Vrms, min (sea level).

Contact resistance (milliohms, max)

Contact	Initial	After
Center	4.0	6 0
Outer	2.0	Not applicable

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Vibration, high frequency: Interruptions - 1 µs, max. Test condition D.

Shock: Test condition I.

Thermal shock: Test condition C.

Moisture resistance: 200 megohms, min.

Corona level: Voltage - 250 V, min.
Altitude - 70,000 feet, min.

RF high potential withstanding voltage. RF voltage - 670 Vrms, min.
Frequency - 5 MHz, min.

Salt spray (corrosion): Test condition B.

Coupling mechanism retention force: 60 lb, min.

MARKING: As specified in MIL-A-55339.

(A) Part No. M55339/29-30101 without safety wire holes, 30001 with safety wire holes.
M55339/29-40101 without safety wire holes, 40001 with safety wire holes.

NOTE: For logistics purposes, only adapters with safety wire holes will be stocked.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3057-11)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL-A-55339/25A

NAME OF ORGANIZATION AND ADDRESS

HQ, USA CORADCOM
ATTN DRDCO-CM-DM
Fort Monmouth, NJ 07703

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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